Optogenetics

Paper: Karl Deisseroth (2011). Foundational perspective; author manuscript.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

What optogenetics can control and how selective perturbations can test causal circuit hypotheses. Provides terminology rather than a disease-specific feedback objective.

How does it answer the general questions?

Actuation. Genetically targeted light-sensitive tools enable rapid excitation, inhibition and modulation of defined cells or projections. Tool choice changes temporal precision, light sensitivity and the duration of induced excitability changes (PDF pp. 1–5, 9).

Measurement and causality. Electrical recordings or optical readouts must separately establish the resulting neural activity. Selective interventions can test whether a cell class or projection contributes to a phenotype, whereas observing its activity alone cannot establish that contribution. The article also stresses that reliable targeting and validated opsin function matter (PDF pp. 2–6).

Model relevance. Supports selecting where u enters a circuit and checking its sign and dynamics experimentally. It gives no unique light-to-firing gain, delay, sigmoid slope or measurement matrix for the notebook. Millisecond-scale capability should not be treated as a universal actuator time constant. No beta-suppression objective or PIE multiplier design follows from this perspective.

What papers does it cite that could provide more information?

- Boyden et al. (2005), *Millisecond-timescale, genetically targeted optical control of neural activity*: foundational neuronal actuation (ref. 4, PDF pp. 6–7).
- Gradinaru et al. (2009), *Optical deconstruction of parkinsonian neural circuitry*: circuit-specific Parkinsonian interventions (ref. 19, PDF p. 7).